

TacDes: Design Interface toward Ubiquitous Tactile Sensing

Xinning He*
Tsinghua University
Beijing City, China
418459483@qq.com

Shiman Zhang*
Edinburgh College of Art
University of Edinburgh
Edinburgh, United Kingdom
s.zhang-224@sms.ed.ac.uk

Luyao Ma
School of Industrial Design
Georgia Institute of Technology
Atlanta, Georgia, USA
lma306@gatech.edu

Jingyi Li
Pomona College
Claremont, California, USA
jingyi.li@pomona.edu

Yiyue Luo
ECE
University of Washington
Seattle, Washington, USA
yiyueluo@uw.edu

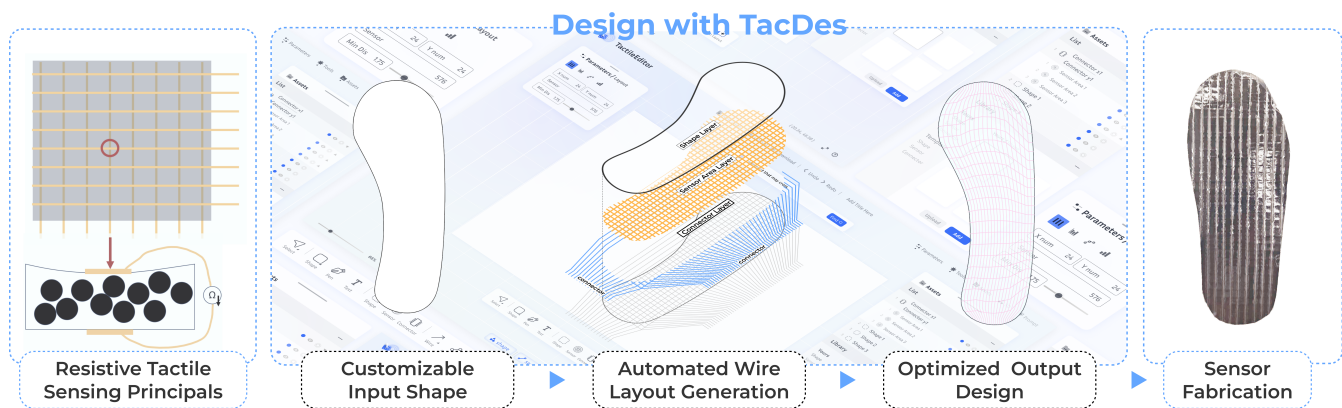


Figure 1: TacDes is a web-based design toolkit to improve the efficiency and accessibility of resistive tactile sensing design. It allows users to import arbitrary geometries, interactively define sensing areas, automatically generate sensing matrix and connecting traces, and export digital files for immediate fabrication.

Abstract

Tactile sensors capture the sensation of touch, providing a unique modality of information for diverse applications in human-computer interaction. However, the complexity and need for customization in tactile sensing interfaces limit their accessibility and potential for broader community. This paper introduces TacDes, a web-based toolkit to streamline the design process of resistive tactile sensors for users at various expertise levels. By performing a formative study with experts and non-experts in tactile sensing, we identify motivations and workflows for tactile sensing interface design. With such insights, we create an open-sourced design toolkit, which facilitates the standardized design process with flexible asset management, automated sensor layout generation, ready-to-use templates, safety warning assistance, and layered exports. Future

work will extend TacDes to 3D designs and evaluate its impact on fabrication and user experience.

CCS Concepts

• **Human-centered computing** → **User interface toolkits; Ubiquitous and mobile computing systems and tools.**

Keywords

tactile sensing, toolkit, interaction design

ACM Reference Format:

Xinning He, Shiman Zhang, Luyao Ma, Jingyi Li, and Yiyue Luo. 2025. TacDes: Design Interface toward Ubiquitous Tactile Sensing. In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems (CHI EA '25)*, April 26–May 01, 2025, Yokohama, Japan. ACM, New York, NY, USA, 8 pages. <https://doi.org/10.1145/3706599.3719721>

1 introduction

A human's sense of touch encodes and conveys a vast amount of information about how we engage with the physical world [17, 53]. Tactile sensors capture this sensation, enriching our ability to interpret and enhance physical interactions [3, 44]. These sensors hold significant value across diverse applications. In human-computer

*Both authors contributed equally to this research.

interaction, touch is one of the most intuitive modalities for engaging with technology [7, 16, 31]. In robotics, capturing human touch interactions with objects and tasks is critical for advancing imitation learning [27, 33, 43]. In healthcare, recording tactile interactions is essential for assessing patient health and identifying issues [11, 35, 59]. In education, teachers benefit from experiencing students' sensation of touch to enhance knowledge transfer [8, 15, 38].

Due to their wide-ranging utility, tactile sensing interfaces must be designed in diverse shapes, sizes, sensor layouts, and configurations to suit specific applications. Developing these highly customized interfaces is often time-intensive and requires advanced expertise in hardware and software, posing significant barriers for non-experts such as hobbyists or researchers from other disciplines. Even for experts, tailoring tactile sensing interfaces to specific usage scenarios involves repetitive, complex, and manual iterations. These challenges hinder the accessibility and widespread adoption of tactile sensing technologies, limiting their potential to achieve ubiquitous application.

To this end, we propose an interactive end-to-end tactile sensing design toolkit to enhance the efficiency and accessibility of tactile sensing design. We first conducted a formative study via a semi-structured interviews with participants from diverse backgrounds to gain a deeper understanding of the sensing design needs and parameter considerations across different user groups. Drawing on the quantitative data and qualitative insights from this study, we developed a web-based tactile sensing design toolkit. The toolkit streamlines repetitive tasks with pre-defined templates and interactive editing, enabling automated sensor layout and connecting traces generation as well as design refinement, enhancing efficiency for experts through standardized workflows and comprehensive parameter selection. It is also designed to be accessible to non-experts, empowering users with a little or no prior hardware experience to engage with tactile sensing design tools more easily. This paper contributes:

- A formative study exploring user needs and design parameters for tactile sensing systems along with quantitative and qualitative analysis.
- The design and implementation of an open-source, web-based interactive tactile sensing design toolkit.
- Demonstrations showcasing the toolkit's capabilities and applications.

2 related work

In this section, we describe tactile sensing mechanisms and the design of interactive toolkits for user-centric applications.

2.1 Tactile Sensing Interfaces

Tactile sensing interfaces capture and interpret physical interactions, primarily associated with touch [17, 28, 29]. They can be classified by their sensing principle, with optical, capacitive, and resistive methods among the most commonly implemented. Optical or vision-based tactile sensors detect touch by analyzing changes in light patterns or images caused by the deformation of a material in response to pressure [22, 36, 50, 60]. While these vision-based methods can be high-resolution and performant, they often require

proprietary hardware and have a fundamental limit in their ability to scale to a large coverage area and a flexible form factor. To this end, capacitive and resistive-based sensing approaches have been more widely deployed for ubiquitous and wearable sensing applications. In capacitive methods, two conductive electrodes typically sandwich an insulating dielectric layer. Applied pressure deforms the dielectric layer and subsequently increases the measured capacitance between the paired electrodes [9, 14, 23, 40, 45, 54]. Similarly, in resistive-based methods, the electrodes sandwich a piezoresistive material instead, and the applied pressure is measured through the change in resistance [5]. In both methods, the electrodes on the top and bottom layers are commonly repeated row-wise and column-wise in an orthogonal fashion to form a two-dimensional "sensing matrix" [21]. The matrix-based sensing approach allows efficient multiplexed signal collection, reducing wiring complexity while ensuring high fidelity. When combined with advancements in fabrication [1, 26, 37, 57], it has allowed resistive and capacitive sensing to be incorporated in a variety of form factors like gloves [24, 51], carpets [25, 56], and pants [49]. This project aims to formulate comprehensive design, firmware, and optimization toolkits for large-area, matrix-based tactile sensing interfaces.

2.2 Interactive Toolkit Design

Toolkit research in HCI addresses the need for comprehensive resources[34] by providing platforms to create interactive artifacts, access complex algorithms, and prototype software and hardware interfaces [19]. Toolkits play a crucial role in reducing authoring time and complexity[12, 32], and lowering the entry barrier by encapsulating low-level knowledge[4, 52]. Application-oriented toolkits are often designed based on the researchers' experiences[30, 47], relevant theories[2] and design considerations[6, 42]. Researchers also conducted formative studies to explore specific user challenges[58]. For example, EIT-kit[61] and SensorViz[18] conducted interviews with experienced and novice users respectively, for designing a toolkit tailored to user-specified resolutions. Toolkits also address limitations of existing design platforms[10, 13, 20, 39, 55, 58]. Our toolkit aims to reduce the complexity and improve the efficiency of resistive tactile sensor design. We perform formative interviews with both expert and non-expert users to understand the challenges in the design process and compiled design principles based on these insights.

3 Formative User Study

To better understand challenges in tactile sensor design and users' needs, we conducted a semi-structured formative study by interviewing both experts and non-experts users. We identify features that make tactile sensor design more accessible and user-friendly, promoting broader adoption and effective utilization.

Participants. We recruited 11 experts in tactile sensor design (9 males, 2 females, average age of 27, average research experience of 3.9 years, denoted as P1-P11), and 5 non-experts with no direct experience in this domain (1 male, 4 females, average age of 25, average research experience of 3.6 years, denoted as NP1-NP5). The experts group includes both experts in resistive tactile sensor design and fabrication (N=5) and experts in related application areas, e.g., robotics and human-computer interactions, with experience

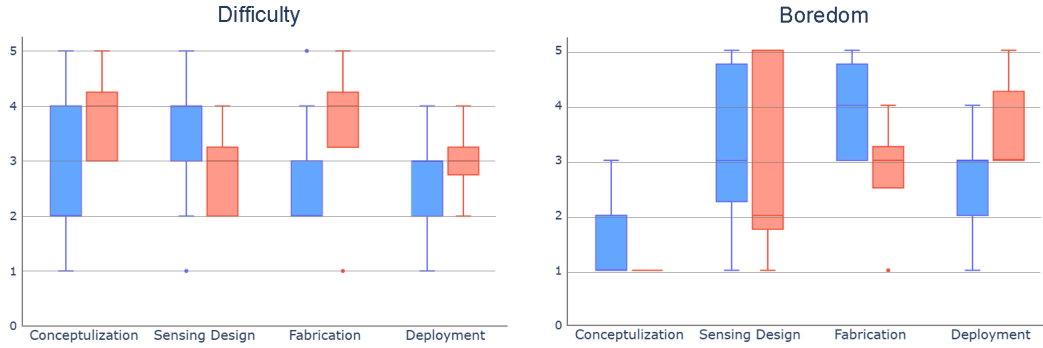


Figure 2: Ratings on the difficulty and boredom of different stages in tactile sensing interfaces development. Ratings from expert and non-expert users are denoted in blue and red respectively.

in resistive tactile sensing (N=6). Non-experts self-reported their knowledge in tactile sensing (avg. 2.4), computer-aided design (avg. 4.4), and fabrication (avg. 4.4) on a scale from 1 to 5.

Procedure. The formative study began by clarifying our scope in resistive tactile sensor designs [25, 48, 51], explaining its mechanisms and potential applications. We used visual materials to explain the process. Participants were then asked to rank the difficulty and boredom for each given working pipeline (concept design, sensing design, fabrication, and user testing) on a scale from 1 to 5 (with 5 indicating the most difficult or boring).

Subsequently, we presented two hypothetical tasks of a 2D and a 3D interactive tactile sensing interface. Experts were instructed to think aloud while solving each task, detailing their thought processes and necessary parameters. Non-experts were asked to refer to the previous process and describe how they expected to complete the tasks based on the given workflow. They were encouraged to share their thoughts during such process. We asked the participants to summarize potential challenges and concerns. Finally, we presented our vision toward such toolkit and asked participants to envision usage frequency, time savings, and effectiveness.

Result. Among all stages of developing a tactile sensing interface, experts rated the design stage as the most difficult (3.45) and the most tedious (3.36). Experts' ratings were primarily based on their development experiences and personal interests. For instance, P5 assigned a high score of 5 for boredom in sensing, stating "It's too easy for me, so I don't find it enjoyable.", while P3 rated a score of 1 as he enjoys "every stage and any kinds of hardware development". Non-expert users found sensing design boring but viewed the stages they were familiar with as more challenging, having experienced the difficulties firsthand. Some non-experts found tactile sensors designs extremely challenging due to a lack of specialized knowledge. They rated based on their hardware-related (NP4) or other collaborative research experiences (NP1, NP2).

The formative study revealed clear distinctions in the needs and motivations for a tactile sensing design toolkit from users at different expertise levels. Experts find the sensing design process tedious and repetitive, and non-experts struggle with technical constraints, particularly understanding the wire overlay processes. Both groups expressing strong motivation for an automatic design pipeline. Additionally, non-experts expressed challenges in collaborative design

setting, highlighting the absence of effective visualization (NP1) and difficulties in aligning functional requirements with technical parameters (NP1, NP5). NP1 described an example for the gap, "If I want different designs, I want to simply choose the ones I feel most suitable from a few preset options. However, the technical team keeps discussing specific parameter changes, which I don't fully understand."

When asked to perform the hypothetical tasks and consider design workflow, experts seek tools to comprehensively evaluate all constraints and requirements when confirming parameters (P6, P11), particularly regarding wire spacing (P1) and alignment (P3, P4, P7, P8) during sensor design. Automated recommendations are also suggested to streamline the iterative layout attempts for unconventional patterns (P9). P2 emphasized the needs for standardized procedures, noting, "Although I am familiar with the operation method, if the entire process is not standardized, it is difficult to replicate over time." Non-expert users, lacking knowledge of parameter settings and wiring methods, expressed a strong preference for automatic generation options (all participants). They highlighted concerns about safety (NP3-NP5), and suggested the inclusion of a scoring metric to quickly evaluate the effectiveness, such as tidiness and resource redundancy (NP1, NP4). Overall, participants proposed high predictions for usage frequency (avg. 8.3/10) and adequate scores (avg. 4.3/5) of the proposed toolkit. However, resistive experts suspected that its accuracy might be limited for complex professional tasks. Experts from other fields preferred off-the-shelf sensors for less personalized tasks. Non-experts felt they could handle simple wiring themselves after the instructions but still preferred automated design generation.

Based on the interviews, we identified two key design principles for our tactile sensing toolkit. First, it should offer standardized workflows and ready-to-use templates to streamline repetitive tasks, improving efficiency and reducing manual effort. Second, it should provide flexible parameters and dynamic layouts to accommodate diverse project needs, allowing experts to make fine-tuned adjustments while guiding beginners through a simplified, stage-by-stage process. Our design and implementation follows these insights.

4 Toolkit Design and Implementation

Our toolkit supports users across different expertise levels to design resistive tactile sensors based on their needs. A typical resistive sensing interface [51] leverages a matrix-like structure, with rows

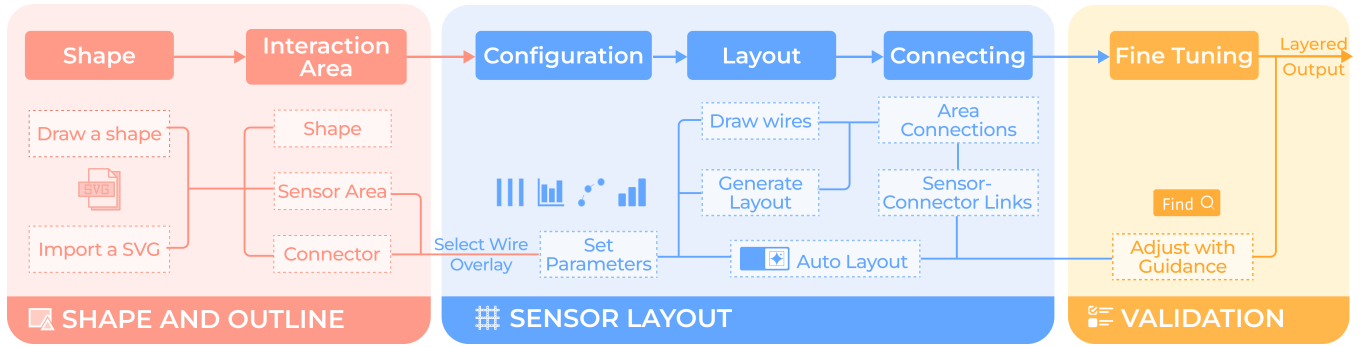


Figure 3: Work flow of the web-based design toolkit.

and columns of electrodes sandwiching a resistive layer. Specifically, we care about the overall shape of the tactile sensing interfaces, arrangement of electrodes, and the extended connecting traces to the circuits. Based on the formative study, we identified three key stages for parameter input and user flow, as shown in Figure 3. First, users can import or define the shape and outline of the tactile sensors. Next, they can generate the sensor layout by specifying parameters such as sensor density and connector positions. This includes generating two sets of overlaying wires inside of the outline as the sensing matrix, as well as the connecting wires that extends to the connector. Finally, the toolkit identifies potential design artifacts, provides warnings, and allows users to export the finalized designs for immediate fabrication.

To support the user workflow, we translate the identified stages into core functional modules integrated in the design toolkit (Figure 4): assets management, automated sensor layout generation, and the guidance and warning system. Each module is detailed below.

4.1 Assets Management

Our toolkit enables users to manage vector assets, with a defined asset framework consisting of *Shapes*, *Sensor Areas*, and *Connectors*, which closely mirrors the workflow of our design toolkit (Fig 5a). *Shapes* define the basic outline, *Sensor Areas* correspond to the wire overlay that forms the sensing matrix, and *Connectors* represent the connection of wires along each direction, which will later interface with the readout circuit for signal serialization. Users can convert the basic shapes into sensor area or connector types, thereby assigning new properties to the vector asset. By defining the subordinate relationships between sensor areas and connectors, the system establishes a connection logic for multiple sensor areas. As all other asset types are derived from the vector-based asset, i.e., SVG, graphics on the canvas share a unified interaction logic, which simplifies operations and reduces the learning curve for users. Additionally, the asset framework supports exporting results in layers, facilitating subsequent fabrication processes.

4.2 Automated Sensor Layout Generation

This module allows users to automatically generate sensor layouts tailored to their needs by selecting from two sensor layout generation approaches and various customizable design parameters.

Moreover, users can define connections between sensor areas, or between sensor areas and connectors. These are achieved with three core components: preprocessing of shapes, grid generation for sensing matrix, and connection algorithms between sensors.

Shape Preprocessing. Shape preprocessing standardizes and optimizes the input geometry, making it suitable for subsequent operations. The shapes handled by users are primarily categorized into curved shapes and polygons. Curved shapes pose greater challenges in processing due to their complex geometry, whereas polygons are easier to handle in vector graphics. To address this, we apply a sampling algorithm to approximate curved shapes as polygons, ensuring consistency and simplifying subsequent processing steps.

Grid Generation Algorithms. The grid generation algorithm subdivides the sensor areas into configurable grids, generating wire traces for sensing matrix within the defined regions. To meet diverse user needs, we provide two types of grids: *Regular Grids* and *Shape-Adapting Grids*. These options allow users to flexibly adapt the grid structure based on the specific requirements of their projects. *Regular Grids* offer a uniform, Cartesian layout that is straightforward and efficient to generate, while *Shape-Adapting Grids* conform to the boundaries of the shapes, enabling precise grid placement for irregular geometries. The density of these grids is customizable by the user, allowing them to adjust the resolution according to their specific requirements.

The regular grid generation process is initiated by determining the bounding box of the target shape, which serves as the spatial constraint for the grid (Fig. 5b-ii). Based on user-defined parameters such as the desired number of divisions or the spacing between grid lines, a uniform grid is created within the bounding box (Fig. 5b-iii). The generated grid lines are then intersected with the boundaries of the shape, effectively slicing the grid to match the shape's dimensions (Fig. 5b-iv). Finally, the intersected segments are used to construct a new grid that fits precisely within the specified area while maintaining the uniform characteristics of a regular grid (Fig. 5b-v).

We also implement a shape-adapting and scan-based grid generation algorithm to distribute a limited number of wires evenly across the arbitrary sensor area. This approach adapts wire orientation to the shape, avoiding clutter or voids and maximizing

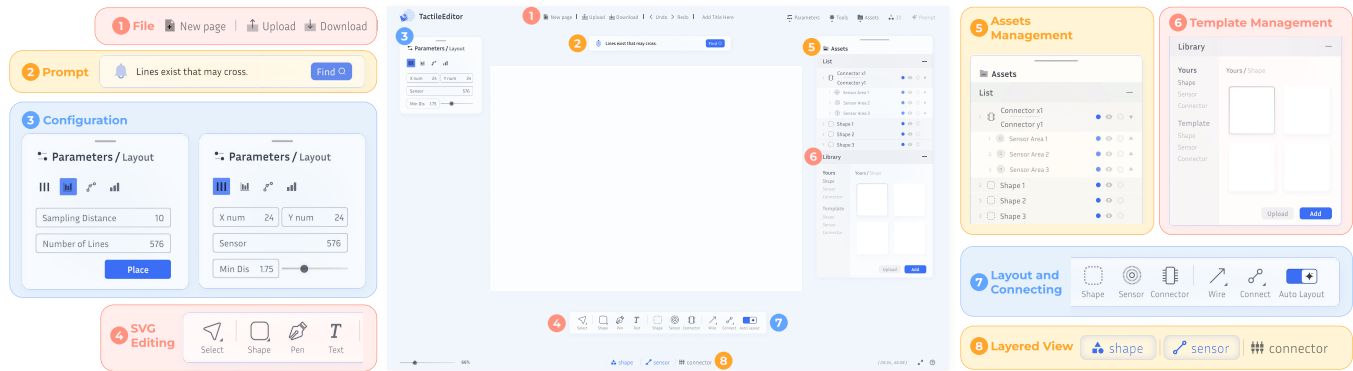


Figure 4: Core functional modules of the design toolkit.

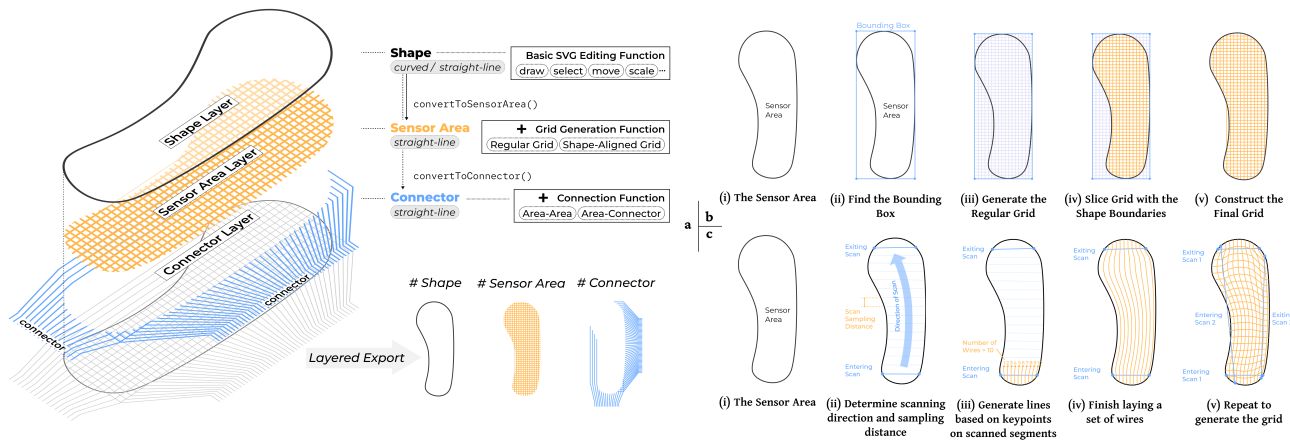


Figure 5: Asset Classification Framework and Grid Generation Algorithms for tactile sensor design: a. Asset Classification Framework and Layered Export structure, illustrating the separation of the design into *Shapes*, *Sensor Areas*, and *Connectors* layers to support layered export and algorithm implementation. b. Regular Grid Generation algorithm. c. Shape-Aligned Grid Generation algorithm.

coverage of the sensor layout in more complex shapes. The algorithm allows users to define the entry and exit points for the wires, ensuring that the wire orientation adapts to the curvature of bent or elongated shapes. The users first define a continuous enclosed area and draw entry and exit segments on the shape's contour, which determine the orientation of the wires (Fig. 5c-i). The scan segments are then placed between the entry and exit points, shifting angles and positions evenly across the shape (Fig. 5c-ii). The scan sampling distance, specified by the user, controls the spacing between adjacent segments, ensuring a smooth and accurate wire placement. Users also define the number of wires that are evenly spaced along each scan segment. These points are connected to the corresponding points on adjacent segments, creating the wire layout (Fig. 5c-iii). This process continues until the scan reaches the exit segment (Fig. 5c-iv), and additional entry-exit pairs can be used to add wires in other directions, allowing a comprehensive and flexible wire distribution (Fig. 5c-v).

Templates. In the toolkit, we provide several pre-designed templates that users can directly select from to simplify their tasks. Users can also save their own designs in the library for future use. These examples also serve as validation for the functionality of our toolkit. Based on previous research on tactile sensor applications [35, 41, 46], we identified three common types of sensors. These include sensors for regular surfaces, e.g., basic squares (Fig. 6.a), circles (Fig. 6.b), and trapezoids (Fig. 6.c), wearables, e.g., shoe soles, socks (Fig. 6.d) and finger cots (Fig. 6.e), as well as robot skin, e.g., an expanded surface for a robotic arm (Fig. 6.f).

Connection Algorithms. Our toolkit allows users to connect multiple sensing areas to minimize the number of wires and simplify the designs. This also allows users to extend the wires to an external connector for piratical applications. During the connection operations, the system automatically identifies selectable intersection points along the contours of the sensor areas and detects available connecting points. After the user selects a pair of intersection point

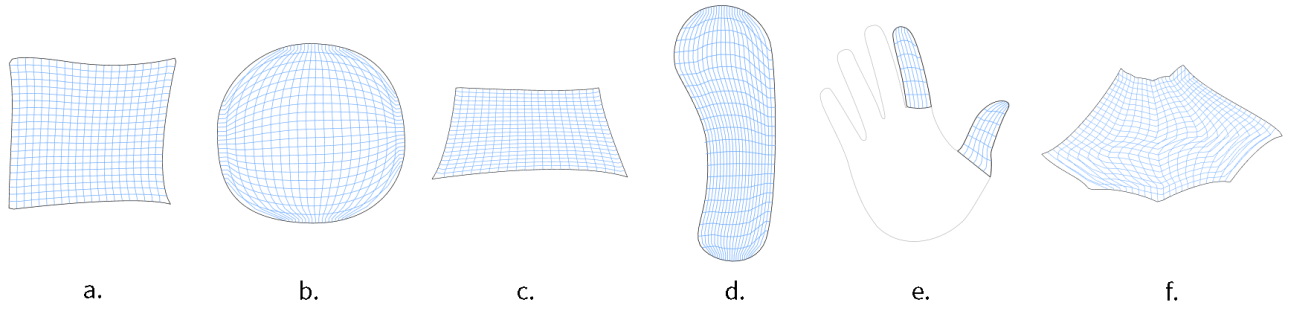


Figure 6: Templates for tactile sensing applications include a-c. surfaces shaped as squares, circles, and trapezoids, d. shoe soles, e. finger cots, f. robot arm coverings.

groups, the algorithm calculates all possible connection sequences and their corresponding intersection points. It then determines the optimal connection order by minimizing the number of intersection points and generates the corresponding connection lines.

4.3 Prompting and Export

This system generates notifications when potential errors are detected to support reliability and robustness of the designs. These notifications assist users in identifying and addressing error points efficiently. We design prompts to address two key aspects, physical performance and accuracy. For example, the system provides alerts when wires intersect or are placed too close to each other, ensuring that the circuit can operate correctly under real-world conditions. The system also prompts users when sensors, i.e., intersection of wires, are detected outside the defined sensor areas, ensuring that the design adheres to the intended layout. The toolkit supports the export of final layout design in a layered format based on the underlying data structure. The exported layers include the outer contour shapes, individual traces for sensing matrix, and a combined layer for connecting traces and connectors. It provides a clear and organized structure. This layered export facilitates seamless integration with fabrication workflows, ensuring that all components of the design are accurately represented for subsequent physical implementation.

Our toolkit and templates are available for open access to support future work in a broader HCI community¹.

5 Discussion and Future Work

User-Centered Evaluation of the Toolkit. In this research, we demonstrated the design and implementation of our tactile sensor design toolkit. In future work, we plan for a comprehensive evaluations of its effectiveness. We will perform user study to thoroughly assess its usability, considering both expert and non-expert user scenarios. We will design comparative experiments for expert users, observing their design process for real-world tasks with and without the toolkit. We will also organize a design workshop for non-experts, to collect quantitative and qualitative feedback from their collaborative design process using the toolkit. We will also validate the

tactile sensor designs via digital fabrication, e.g., laser cutting, 3D printing, and digital embroidery.

Extending the Toolkit for 3D Designs. We have identified that 3D tactile sensors design posed greater complexity. The current research focused exclusively on 2D design scenarios, but it also provides great flexibility for future functionality integration. We plan to extend our algorithm and toolkit development to support 3D design tasks by enabling the import of STL files and performing UV surface segmentation and unfolding, with clear visualizations of how the plain wire layout corresponds to the model.

Integrating the Toolkit with Prompts. In the formative study, both experts and non-experts expressed a desire for natural language interaction or an automatic, one-click design completion feature. We plan to explore the integration of Large Language Models (LLMs) to better understand and translate real-world requirements to comprehensive, automated recommendations for sensing design parameters. Additionally, we plan to integrate the toolkit with safety checks and provide intuitive visualized warnings about potential risks for non-experts.

6 Conclusion

In summary, we present the design and implementation of an open-source, web-based interactive toolkit that facilitates resistive tactile sensing design. We conducted a formative study to extract the motivations and design considerations from participants with varying knowledge backgrounds. Based on both quantitative and qualitative analysis, we designed and implemented the design toolkit with the core functions of asset management, sensor layout generation, and prompting and export. Our demonstrations highlight the capability of our toolkit, demonstrating potential for ubiquitous tactile sensing applications.

Acknowledgments

This work is supported by the 2024 Access Computing Summer Program, which is sponsored by the Global Innovation Exchange (GIX), a global partnership between Tsinghua University and the University of Washington.

¹<https://github.com/zsmmmmer/TacDes>

References

- [1] Roland Aigner, Andreas Pointner, Thomas Preindl, Patrick Parzer, and Michael Haller. 2020. Embroidered Resistive Pressure Sensors: A Novel Approach for Textile Interfaces. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (CHI '20). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376305
- [2] Giulia Barbareschi, Enrico Costanza, and Catherine Holloway. 2020. TIP-Toy: a tactile, open-source computational toolkit to support learning across visual abilities. In *Proceedings of the 22nd international ACM SIGACCESS conference on computers and accessibility*. 1–14.
- [3] Chiara Bartolozzi, Lorenzo Natale, Francesco Nori, and Giorgio Metta. 2016. Robots with a sense of touch. *Nature materials* 15, 9 (2016), 921–925.
- [4] Patrick Baudisch, Stefanie Mueller, et al. 2017. Personal fabrication. *Foundations and Trends® in Human-Computer Interaction* 10, 3–4 (2017), 165–293.
- [5] Bijender and Ashok Kumar. 2022. Recent progress in the fabrication and applications of flexible capacitive and resistive pressure sensors. *Sensors and Actuators A: Physical* 344 (2022), 113770. doi:10.1016/j.sna.2022.113770
- [6] Hans Brombacher, Rosa Van Koningsbruggen, Steven Vos, and Steven Houben. 2024. SensorBricks: a Collaborative Tangible Sensor Toolkit to Support the Development of Data Literacy. In *Proceedings of the Eighteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–17.
- [7] Alex Butler, Shahram Izadi, and Steve Hodges. 2008. SideSight: multi-“touch” interaction around small devices. In *Proceedings of the 21st annual ACM symposium on User interface software and technology*. 201–204.
- [8] Asta Cekaite and Disa Bergnehr. 2018. Affectionate touch and care: Embodied intimacy, compassion and control in early childhood education. *European early childhood education research journal* 26, 6 (2018), 940–955.
- [9] Wufan Chen and Xin Yan. 2020. Progress in achieving high-performance piezoresistive and capacitive flexible pressure sensors: A review. *Journal of Materials Science & Technology* 43 (2020), 175–188. doi:10.1016/j.jmst.2019.11.010
- [10] Artem Dementyev, Hsin-Liu Kao, and Joseph A Paradiso. 2015. Sensortape: Modular and programmable 3d-aware dense sensor network on a tape. In *Proceedings of the 28th Annual ACM Symposium on User Interface Software & Technology*. 649–658.
- [11] Lazzaro Di Biase, Alessandro Di Santo, Maria Letizia Caminiti, Alfredo De Liso, Syed Ahmar Shah, Lorenzo Ricci, and Vincenzo Di Lazzaro. 2020. Gait analysis in Parkinson’s disease: An overview of the most accurate markers for diagnosis and symptoms monitoring. *Sensors* 20, 12 (2020), 3529.
- [12] Ellen Yi-Luen Do and Mark D Gross. 2007. Environments for creativity: a lab for making things. In *Proceedings of the 6th ACM SIGCHI conference on Creativity & cognition*. 27–36.
- [13] Martin Feick, Cihan Biyikli, Kiran Gani, Anton Wittig, Anthony Tang, and Antonio Krüger. 2023. VoxelHap: A toolkit for constructing proxies providing tactile and kinesthetic haptic feedback in virtual reality. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–13.
- [14] Tobias Grosse-Puppenthal, Yannick Berghoefer, Andreas Braun, Raphael Wimmer, and Arjan Kuijper. 2013. OpenCapSense: A rapid prototyping toolkit for pervasive interaction using capacitive sensing. In *2013 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 152–159. doi:10.1109/PerCom.2013.6526726
- [15] Kevin Huang, Thad Starner, Ellen Do, Gil Weinberg, Daniel Kohlsdorf, Claas Ahlrichs, and Ruediger Leibbrandt. 2010. Mobile music touch: mobile tactile stimulation for passive learning. In *Proceedings of the SIGCHI conference on human factors in computing systems*. 791–800.
- [16] Hiroshi Ishii. 2008. Tangible bits: beyond pixels. In *Proceedings of the 2nd international conference on Tangible and embedded interaction*. xv–xxv.
- [17] Roland S Johansson and J Randall Flanagan. 2009. Coding and use of tactile signals from the fingertips in object manipulation tasks. *Nature Reviews Neuroscience* 10, 5 (2009), 345–359.
- [18] Yoonji Kim, Junyi Zhu, Mihir Trivedi, Dishita Turakhia, Ngai Hang Wu, Donghyeon Ko, Michael Wessely, and Stefanie Mueller. 2022. SensorViz: Visualizing Sensor Data Across Different Stages of Prototyping Interactive Objects. In *Proceedings of the 2022 ACM Designing Interactive Systems Conference*. 987–1001.
- [19] David Ledo, Steven Houben, Jo Vermeulen, Nicolai Marquardt, Lora Oehlberg, and Saul Greenberg. 2018. Evaluation strategies for HCI toolkit research. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–17.
- [20] Yuxuan Lei, Qi Lu, and Yingqing Xu. 2022. O&O: A DIY toolkit for designing and rapid prototyping olfactory interfaces. In *Proceedings of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–21.
- [21] Pierre Lemerle, A Klinger, Alessandra Cristalli, and M Geuder. 2008. Application of pressure mapping techniques to measure push and gripping forces with precision. *Ergonomics* 51 (03 2008), 168–91. doi:10.1080/00140130701528602
- [22] Nathan F Lepora and Benjamin Ward-Cherrier. 2015. Superresolution with an optical tactile sensor. In *2015 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*. IEEE, 2686–2691.
- [23] Ruiqing Li, Qun Zhou, Yin Bi, Shaojie Cao, Xue Xia, Aolin Yang, Siming Li, and Xueliang Xiao. 2021. Research progress of flexible capacitive pressure sensor for sensitivity enhancement approaches. *Sensors and Actuators A: Physical* 321 (2021), 112425. doi:10.1016/j.sna.2020.112425
- [24] Yichen Li, Yilun Du, Chao Liu, Chao Liu, Francis Williams, Michael Foshey, Benjamin Eckart, Jan Kautz, Joshua B. Tenenbaum, Antonio Torralba, and Wojciech Matusik. 2024. Learning to Jointly Understand Visual and Tactile Signals. In *The Twelfth International Conference on Learning Representations*. https://openreview.net/forum?id=NtQqIcSbqv
- [25] Yiyue Luo, Yunzhu Li, Pratyusha Sharma, Wan Shou, Kui Wu, Michael Foshey, Beichen Li, Tomás Palacios, Antonio Torralba, and Wojciech Matusik. 2021. Learning human–environment interactions using conformal tactile textiles. *Nature Electronics* 4, 3 (2021), 193–201.
- [26] Yiyue Luo, Kui Wu, Tomás Palacios, and Wojciech Matusik. 2021. KnitUI: Fabricating Interactive and Sensing Textiles with Machine Knitting. In *Proceedings of the 2021 CHI Conference on Human Factors in Computing Systems* (Yokohama, Japan) (CHI '21). Association for Computing Machinery, New York, NY, USA, Article 668, 12 pages. doi:10.1145/3411764.3445780
- [27] Jeffrey Mahler, Matthew Matl, Vishal Satish, Michael Danielczuk, Bill DeRose, Stephen McKinley, and Ken Goldberg. 2019. Learning ambidextrous robot grasping policies. *Science Robotics* 4, 26 (2019), eaau4984.
- [28] Margaret W Matlin and Hugh J Foley. 1992. *Sensation and perception*. Allyn & Bacon.
- [29] M-Marsel Mesulam. 1998. From sensation to cognition. *Brain: a journal of neurology* 121, 6 (1998), 1013–1052.
- [30] Kouta Minamizawa, Yasuaki Kakehi, Masashi Nakatani, Soichiro Mihara, and Susumu Tachi. 2012. TECHTILE toolkit: a prototyping tool for design and education of haptic media. In *Proceedings of the 2012 Virtual Reality International Conference*. 1–2.
- [31] Marianna Obrist, Carlos Velasco, Chi Thanh Vi, Nimesha Ranasinghe, Ali Israr, Adrian D Cheok, Charles Spence, and Ponnampalam Gopalakrishnakone. 2016. Touch, taste, & smell user interfaces: The future of multisensory HCI. In *Proceedings of the 2016 CHI conference extended abstracts on human factors in computing systems*. 3285–3292.
- [32] Dan R Olsen Jr. 2007. Evaluating user interface systems research. In *Proceedings of the 20th annual ACM symposium on User interface software and technology*. 251–258.
- [33] Luke E Osborn, Andrei Dragomir, Joseph L Betthausen, Christopher L Hunt, Harrison H Nguyen, Rahul R Kaliki, and Nitish V Thakor. 2018. Prosthesis with neuromorphic multilayered e-skin perceives touch and pain. *Science robotics* 3, 19 (2018), eaat3818.
- [34] Antti Oulasvirta and Kasper Hornbæk. 2016. HCI research as problem-solving. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*. 4956–4967.
- [35] Oliver Ozioko and Ravinder Dahiya. 2022. Smart tactile gloves for haptic interaction, communication, and rehabilitation. *Advanced Intelligent Systems* 4, 2 (2022), 2100091.
- [36] Akhil Padmanabha, Frederik Ebert, Stephen Tian, Roberto Calandra, Chelsea Finn, and Sergey Levine. 2020. OmniTact: A Multi-Directional High-Resolution Touch Sensor. In *2020 IEEE International Conference on Robotics and Automation (ICRA)*. 618–624. doi:10.1109/ICRA40945.2020.9196712
- [37] Patrick Parzer, Florian Perteneder, Kathrin Probst, Christian Rendl, Joanne Leong, Sarah Schuetz, Anita Vogl, Reinhard Schwoedauer, Martin Kaltenbrunner, Siegfried Bauer, and Michael Haller. 2018. RESI: A Highly Flexible, Pressure-Sensitive, Imperceptible Textile Interface Based on Resistive Yarns. In *Proceedings of the 31st Annual ACM Symposium on User Interface Software and Technology* (Berlin, Germany) (UIST '18). Association for Computing Machinery, New York, NY, USA, 745–756. doi:10.1145/3242587.3242664
- [38] Heather Piper and Hannah Smith. 2003. ‘Touch’ in educational and child care settings: dilemmas and responses. *British educational research journal* 29, 6 (2003), 879–894.
- [39] Dominic Potts, Martynas Dabravalskis, and Steven Houben. 2022. TangibleTouch: A Toolkit for Designing Surface-based Gestures for Tangible Interfaces. In *Proceedings of the Sixteenth International Conference on Tangible, Embedded, and Embodied Interaction*. 1–14.
- [40] Narjes Pourjafarian, Anusha Withana, Joseph A Paradiso, and Jürgen Steimle. 2019. Multi-Touch Kit: A do-it-yourself technique for capacitive multi-touch sensing using a commodity microcontroller. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology*. 1071–1083.
- [41] Soonjae Pyo, Jaeyong Lee, Kyubin Bae, Sangjun Sim, and Jongbaeg Kim. 2021. Recent progress in flexible tactile sensors for human-interactive systems: from sensors to advanced applications. *Advanced Materials* 33, 47 (2021), 2005902.
- [42] Di Qi, Mina Shibasaki, Youichi Kamiyama, Sakiko Tanaka, Bunsuke Kawasaki, Chisa Mitsuhashi, Yun Suen Pai, and Kouta Minamizawa. 2022. Furekit: Wearable tactile music toolkit for children with ASD. In *International Conference on Human Haptic Sensing and Touch Enabled Computer Applications*. Springer, 310–318.
- [43] Joseph M Romano, Kaijen Hsiao, Günter Niemeyer, Sachin Chitta, and Katherine J Kuchenbecker. 2011. Human-inspired robotic grasp control with tactile sensing.

- IEEE Transactions on Robotics* 27, 6 (2011), 1067–1079.
- [44] Hannes P Saal, Benoit P Delhay, Brandon C Rayhaun, and Sliman J Bensmaia. 2017. Simulating tactile signals from the whole hand with millisecond precision. *Proceedings of the National Academy of Sciences* 114, 28 (2017), E5693–E5702.
 - [45] Valkyrie Savage, Xiaohan Zhang, and Björn Hartmann. 2012. Midas: fabricating custom capacitive touch sensors to prototype interactive objects. In *Proceedings of the 25th annual ACM symposium on User interface software and technology*. 579–588.
 - [46] Yitian Shao, Hui Hu, and Yon Visell. 2020. A wearable tactile sensor array for large area remote vibration sensing in the hand. *IEEE Sensors Journal* 20, 12 (2020), 6612–6623.
 - [47] Mina Shibasaki, Keitaro Tuchiya, and Kouta Minamizawa. 2017. TECHTILE bits: Designing a tool for children to create haptic devices. In *Proceedings of the Virtual Reality International Conference-Laval Virtual 2017*. 1–4.
 - [48] Zilin Si, Tianhong Catherine Yu, Katrene Morozov, James McCann, and Wenzhen Yuan. 2023. RobotSweater: Scalable, generalizable, and customizable machine-knitted tactile skins for robots. In *2023 IEEE International Conference on Robotics and Automation (ICRA)*. IEEE, 10352–10358.
 - [49] Gurashish Singh, Alexander Nelson, Ryan Robucci, Chintan Patel, and Nilanjan Banerjee. 2015. Inviz: Low-power personalized gesture recognition using wearable textile capacitive sensor arrays. In *2015 IEEE International Conference on Pervasive Computing and Communications (PerCom)*. 198–206. doi:10.1109/PERCOM.2015.7146529
 - [50] Huanbo Sun, Katherine J Kuchenbecker, and Georg Martius. 2022. A soft thumb-sized vision-based sensor with accurate all-round force perception. *Nature Machine Intelligence* 4, 2 (2022), 135–145.
 - [51] Subramanian Sundaram, Petr Kellnhofer, Yunzhu Li, Jun-Yan Zhu, Antonio Torralba, and Wojciech Matusik. 2019. Learning the signatures of the human grasp using a scalable tactile glove. *Nature* 569, 7758 (2019), 698–702.
 - [52] Hannah Twigg-Smith, Emily Whiting, and Nadya Peek. 2024. KnitScape: Computational Design and Yarn-Level Simulation of Slip and Tuck Colorwork Knitting Patterns. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–20.
 - [53] Ake B Vallbo, Roland S Johansson, et al. 1984. Properties of cutaneous mechanoreceptors in the human hand related to touch sensation. *Hum neurobiol* 3, 1 (1984), 3–14.
 - [54] Chi Cong Vu, Seung Ju Kim, and Jongwon Kim. 2021. Flexible wearable sensors - an update in view of touch-sensing. *Science and Technology of Advanced Materials* 22, 1 (March 2021), 26–36. doi:10.1080/14686996.2020.1862629
 - [55] Qi Wang, Yuan Zeng, Runhua Zhang, Nianding Ye, Linghao Zhu, Xiaohua Sun, and Teng Han. 2023. EmTex: Prototyping Textile-Based Interfaces through An Embroidered Construction Kit. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–17.
 - [56] Irmandy Wicaksono, Don Derek Haddad, and Joseph Paradiso. 2022. Tapis Magique: Machine-knitted Electronic Carpet for Interactive Choreomusical Performance and Immersive Environments. In *Proceedings of the 14th Conference on Creativity and Cognition (Venice, Italy) (C&C '22)*. Association for Computing Machinery, New York, NY, USA, 262–274. doi:10.1145/3527927.3531451
 - [57] Irmandy Wicaksono, Peter G Hwang, Samir Droubi, Franny Xi Wu, Allison N Serio, Wei Yan, and Joseph A Paradiso. 2022. 3dknits: Three-dimensional digital knitting of intelligent textile sensor for activity recognition and biomechanical monitoring. In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. IEEE, 2403–2409.
 - [58] Chengshuo Xia, Tian Min, Daxing Zhang, and Congsi Wang. 2024. Understanding the Needs of Novice Developers in Creating Self-Powered IoT. In *Proceedings of the CHI Conference on Human Factors in Computing Systems*. 1–17.
 - [59] Rasoul Yousefi, Sarah Ostadabbas, Miad Faezipour, Mehrdad Nourani, Vincent Ng, Lakshman Tamil, Alan Bowling, Deborah Behan, and Matthew Pompeo. 2011. A smart bed platform for monitoring & ulcer prevention. In *2011 4th international conference on biomedical engineering and informatics (BMEI)*, Vol. 3. IEEE, 1362–1366.
 - [60] Wenzhen Yuan, Siyuan Dong, and Edward H. Adelson. 2017. GelSight: High-Resolution Robot Tactile Sensors for Estimating Geometry and Force. *Sensors* 17, 12 (2017). doi:10.3390/s17122762
 - [61] Junyi Zhu, Jackson C. Snowden, Joshua Verdejo, Emily Chen, Paul Zhang, Hamid Ghaednia, Joseph H Schwab, and Stefanie Mueller. 2021. EIT-kit: An electrical impedance tomography toolkit for health and motion sensing. In *The 34th Annual ACM Symposium on User Interface Software and Technology*. 400–413.